

# Interaction of Spatial Computing In Augmented Reality

S. Balakrishnan

Dept. of Computer Science and  
Business Systems, Sri Krishna College  
of Engg. and Technology, Coimbatore,  
Tamilnadu, India

M.Syed Shahul Hameed

University of Technology and Applied  
Sciences - SUR,  
PO BOX : 484, PC 411, SUR, The  
Sultanate of Oman.

Venkatesan. K

St. Josephs college of engineering  
Department of ECE  
venkatsri21@gmail.com

Aswin G

Full stack software developer  
zealbots in chadura Tech,  
Tiruchirapalli,  
Tamilnadu, India.

**Abstract**—In the above certain years, the technology of spatial computing has recognized the environment growth of simulated reality by user interaction and extensive knowledge which assist the 3D canvas used space for user interfaces. Spatial computing is widely compatible with XR which is an Extended Reality. This acts itself as an umbrella phase for Augmented Reality (AR), Mixed Reality (MR) and Virtual Reality (VR). Moreover, we are looking into the deep concepts of AR in spatial computing. Augmented reality is a new medium of laminating and merging digital contents towards the real world. This medium provides individual assets connecting the physical-virtual world with steady control and enhances the overall environment reality of user's information. These are obtained based on mobile devices, projection devices and mounted displays that help in augmenting the information over real-life. In advance, AR is part of spatial augmented reality which is an assistive technology tool used for determining the completion time of task, speed range and quality of interactive computation. This paper delivers the effectiveness of Augmented Reality (AR) with possible combination of technologies and applications from a user's point of view.

**Index Terms:** *Spatial computing, augmented reality, virtual reality, mixed reality.*

## I. INTRODUCTION

“Spatial Computing” is one of the huge concepts of connecting digital and physical reality into the actual world. This is almost done based on Virtual Reality (VR) and Augmented Reality (AR). This assists in digitalizing the object through the cloud, permits the sensors to react with each other and digitalizes the real world representation. Further, this integrates the accomplishment of spatial mapping to validate computer tracking and navigates the interaction process between object and person with physical to digital world. The spatial computing helps in integrating connection between human to machine and machine to machine which process various operations based on classified advanced efficiency level of lives, industries, medical-care, homes and transports. The top companies invested in spatial computing are Amazon and Microsoft technologies. Here, spatial computing technology helps in

developing the simulated environment reality for user interactions. The first phase of the technology is to access Virtual Reality (VR) and involve the user towards 3D dimension, artificial and digital mounted display with specific interaction between users and real world. Further, VR computations are used in many fields for increasing the virtual space in games, entertainment, education and sports. In the medical aspect, this assists in training surgical methods for viewing interventional approaches. The next phase of the technology is to operate Augmented Reality (AR) which fixes and overlays all the digital concepts towards the real-time world. This reality elaborates the information content to the user's conditions. This attains devices like monitoring screening, smartphones and google camera. AR augments in virtual space with information on the real world. The digital world observes, quantifies and manipulates augmented reality for representing numerous technologies [1].

## II. SPATIAL COMPUTING DEEP DIVE

RPA is a “utilization of innovation that is represented Spatial computing used 3D canvas space for user interfaces. Extended Reality (ER) is the broad phase of spatial computing. This holds all types of realities like augment, virtual and mixed which helps in computing XR user interactions. The human brain gives out three dimensional physical conditions to the real world. Here, the languages used for enhancing the computation is based on physical speech for grasping the knowledge. Spatial computing analyses the extensive recognition of information to gain knowledge. In this numerous digital concepts are sorted such as data visualization, story analysis and mathematics contents with a huge number of structures. This explores human intellectual capability and expands the true expectation over physical tasks with certain human abilities. Thus, this enables the concept of collaboration between humans and machines which utilize synchronization processes [2-5].

The three concepts of spatial computing includes:

(i) Enable the technologies to perceive the digital 3D concepts.

Example: Augmented Reality and Virtual Reality headsets

(ii) Allow the technologies to interact with digital 3D concepts.

Example: Voice controlling and Eye tracking

(iii) Effective ideology for 3D UX design.

Audio Visual Technology

Spatial computing turns on with 3D imaging technologies. This includes AR Headsets, Holographic Displays, GearVR, HoloLens, Magic Leap and Looking Glasses. The advancement in the computation helps in generating and solving the puzzles through various techniques. The Augmented-Headset is an exclusive type of head-moduled screen display that gives out simulate visualization over physical environment to digital mode display. The users are allowed to hear and watch the world through a determined headset glass. The Holographic display helps in projecting the graphical image lighting that is used in scenes to view the objects picture and record the pattern such as X-ray images. GearVR is a virtual reality headset which consists of more specific features like touch pads and sensors to detect. It is the combination of Oculus VR which analyses the screen display and obtains outcome imaging. The audio and visual technologies are important components of spatial computing [6].

#### Interaction Technology

The interaction technology of spatial computing is based on three dimensional authentication of visual communications. This assists in digitalizing the object through the cloud, permits the sensors to react with each other and digitalizes the real world representation. Further, this integrates the accomplishment of spatial mapping to validate computer tracking and navigates the interaction process between object and person with physical to digital world. The user interface of spatial computing includes body gestures, eye and voice control connections [7].

The four important interaction technologies are:

| Interaction Technologies | Specifications                                                      |
|--------------------------|---------------------------------------------------------------------|
| Hand Tracking            | Hand tracking automatically interacts and operates virtual objects. |
| Haptics                  | It is virtual touch wearable technology made of augmented reality.  |
| Voice Control            | This is voice based digital assistance engaged with VR and AR.      |
| Eye Tracking             | Eye tracking enhances most effective delivering of visualization.   |

#### 3D UX/UI Design

The potential of spatial computing is determined by unlocked technologies which are used for mirroring the design of 2D-interactions but in the case of 3D interface it needs a UX design for collaborating the connection between human interactions. This materializes the standard Extensive Reality (XR) where more things are to be added. In specific this is genuine interactions. This holds various UX principles practices such as eye/hand tracking, voice controlling and haptics for certain progress of work. The 3D UX includes the (shown in figure 2) interaction design, prototype frames, information architectures, user research and many other scenarios, The 3D UI consists of visual designs, colors, graphic designs, layouts and typography of the real world. This explores human intellectual capability and expands the true expectation over physical tasks with certain human abilities [8].

#### Spatial Computing Intelligent Machines

Spatial Computing uses intelligent machines for enhancing the process of visualization in three-dimension augmented reality. This technology is to operate Augmented Reality (AR) which fixes and overlays all the digital concepts towards the real-time world. This reality elaborates the information content to the user's conditions. This includes Browsers, 3D models and slice tools. The projection of intelligent objects helps in determining the solutions for various things without pressing on keyboard or mouse. These are operated over educational areas, entertainment fields, industries, modelling, games, health-care, space and scientific researchers. In advance, AR is part of spatial augmented reality which is an assistive technology tool used for determining the completion time of task, speed range and quality of interactive computation.

### III. AUGMENTED REALITY

A physical framework with existence of mind in a perceived time state is known as Reality. Nowadays we can also exist inside the computer generated space and time that is known as virtual reality. In between this another kind of reality can be enhanced which includes normal reality and virtual reality that is called "Augmented Reality." AR processes reality with bits of virtual reality. In another perspective this emphasizes the virtual objects into the physical world rather than going around. AR assists in seeing the nature of life before us like trees around the park, dogs following the balls, children playing football they have an augmentation added to digitalization. For example, dinosaurs eating leaves of trees, dogs may be mixed with their cartoon counterparts, and children may be seen kicking past a space shuttle where they will score a goal [9].

Augmented reality (AR) is an important technology movement right now and. This helps in enhancing the growth of AR Smartphones and other devices that are readily available around the world. With the advancement of AR technology, these models can be accessed through different smartphone applications. The augmented reality consists of specific apps such as to identify cars overcrowded on particular location, to capture the lenses on

snapchat and to enhance numerous shopping applications with trail clothes without moving out from home. AR is enveloping an open technology which magnifies virtual items to the real world of greater user's. Augmented Reality holds 25% virtual and 75% real displaying information. In this, the users pursue to hold in touch with virtual and real life interactions. AR devices digitize the items from the physical environment to the top layers of elements. This enhances a computer initiated images in order to sight the visibility of contrasting AR devices. Here, the smart devices are more adaptable to augmented reality for acquiring additional experience. Further, the users have privilege to swap or connect more insights to real-time with lower cost applications on mainstream media. This uses all smartphones, tablets and mobile devices where the system arguments with real world scenes. In fact, while moving in a visual environment, user actions (movements / communication) have a direct effect on a given material and all of this happens in real time. If you pay close attention, these rules do not discriminate against any particular instrument of output or contact area, which means that, in this theory AR is not limited to visual display and performance. However, audio, haptic and AR gustatory are very difficult to achieve.

Properties abided in Augmented Reality:

- (i) This merges virtual and real information with primary actions towards the real world.
- (ii) This has Real-time interactive updates.
- (iii) It contains virtual information on 3D space of the physical environment.

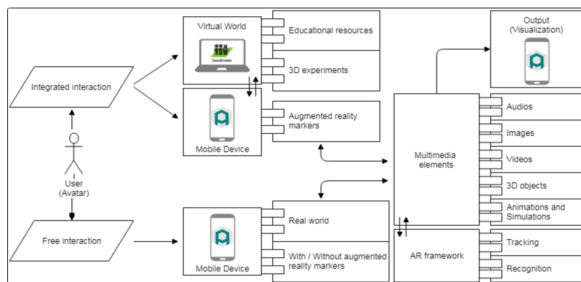


Fig 1: Architecture of Augmented Reality

The figure 1 shows the integrated interaction between the user and augmented reality in order to obtain output over visualization. The collaboration is around virtual world and mobile devices where the virtual environment enhances the resources eg: Education. Further, this experiments 3D augmented reality towards the information of the real world. The multimedia elements and AR frameworks are interlinked. Finally, this process includes audios, images, videos, 3D objects, animations, stimulations, tracking and recognition over visualized information in Augmented reality where this forms up with good architecture.

#### IV. EXISTING TOOLS OF AR

However, portable augmented devices are becoming more and more reliable ways for screening head-mounted displays and it should come as no surprise that in recent

years they have become the most expensive way to improve Augmented Reality and more importantly to make them widely available to the users. These days, almost many of the smartphones can support AR. All you need is a camera, a display, and enough computer power to ensure a "real-time" connection with the given visualization. So as we have seen, AR goes a long way from its origins, and these days it is already doing amazing work. Here, AR already has its own potential and what to expect in the future. Each year additionally AR and VR games are added into the app stores, especially AR-based games as they usually do not require the purchase of additional expensive hardware. Currently, augmented reality progresses to portable power supplies, displays and processors over finite technologies.

The tools used for building and testing augmented reality of apps are [11]:

(i) ARToolKit: This tool is used for tracking the viewpoint and analysing the virtual object interactions. The general platform used by ARToolKit is Windows, Linux, Android and iOS.

(ii) ARKit: This persistent multiple user detection and tracking to detect the 3D objects such as sculpture, furniture and toys. It is mainly designed for iOS users.

(iii) ARCore: This platform is developed for AR experience for motion tracking on camera and understanding the environmental light estimation. This API enables Android and iOS.

#### V. Companies Based on AR

(i) Pepsi Max

This company pranks the Londoner communication and authorizes the Augmented reality display for bus stops while travelling, prevents the metro from clash, identify aliens grabbing peoples which programme to justice.

(ii) US Army

This gives a situational recognition of soldiers of the United States Army with AR based technologies. Here, TAR which is Tactical augmented Reality which is used as tech essential for soldiers eyepiece to mark the location of companion and enemy. The TAR helps in night vision goggles to see the objects at dark mode. This handles a GPS system which determines the exact position of soldiers carrying each and every day to analyze their location if they seem to be missing. A wireless eyepiece is connected over a soldiers tablet which is a wearable device that connects with mounted thermal site rifles. In order the soldier obtained the clear view of the target to be pointed with imaging. Here, almost long distance targets can be identified using eyepiece of Augmented reality technology.

(iii) Disney

This research helped out in coloring books with 3D characters. Augmented Reality (AR) which fixes and overlays all the digital concepts. The children are more initiated to colour up the character of disney world and Moreover, this is included as the example because it is a normal application with lots of potentials towards the AR.

(iv) Acura

It streams live races featuring Augmented reality. This is attached to the wheels of racing bikes in order to influence the new model of Acura. AR race is the most rapid experience of the single running in athletics with three laps shoots up to identify overall time durations. Acura is visible on various phases such as for drivers and live audiences on Facebook. This uniquely finds out the visualization and obstacle of Acura that is to be handled properly. The car drivers mount the cameras over car's and the bike rider manages to attach it on the helmet to analyze the actions going around. The AR technology connects all the additional wide visors to predict the original colour in viewing experience. Here, the helmets are associated with computers to obtain a high performance over rendering process. The Acura maintains the capacity of experience with smooth visuals at high speed.

(v) L' Oreal

This enhances the makeup brand collections into certain makeup apps like YouCam Perfect which elaborates the challenges of customers. In general to discover traditional beauty products. This contains several features such as contouring, colour fixing and strobing. L'Oreal is in partnership with Perfect crops which gives out ipads with beauty contours. This allows the customers to makeup so efficiently and comfortably from home itself. In present 64 makeup apps have been launched over app stores of India, USA and Russia. In addition, this cooperates with collection for products for delivery where the customers can purchase up all the L'Oreal Paris products over app and it is delivered to ordered users through online communication using AR developed technology.

(vi) Bonus

This observes the view of Google videos and predicts the recommended search technology for immense process of work. This explores human intellectual capability and expands the true expectation over physical tasks. The Bonus consists of AR and VR which are more immersed in our regular lives.

## VI. BIG APPLICATIONS

(i) Airport application

The creative usage of AR technology in airports has immense enhancement. This is an airport passenger application which won lots of awards. Gatwick has developed this application and this helped more than 2,000 beacons all around the extreme terminal. This mobile application assists the passenger to trace out the nearby airport station locations and supports in reducing more traffic movements on airports.

(ii) Applications of remodeled and Dulux Visualiser

In this, purchasing furniture and delivering on time was a greater discovery but this did not enhance the work of specific locations. They developed an application named Ikea which elaborates identification of required places to avoid future clashing. This application was constructed by ARkit of Apple technology which recognizes the space plan and analyzes the Ikea items on the specific room that

digitizes the images of the latest products. Further, they deployed a Lowe's store to estimate the virtual tape application which detailed 3D image items. This furnishes attachment in profit-oriented home space. The duplex visualiser assists in buying a compatible room with various paint shades using smartphone camera technology which finally examines the paint and virtual room with VIBGYOR colours.

(iii) Virtual applications of Virtual Artist

An augmented reality technology was developed over a cosmetic company. The name of the company is Sephora which inspects the client with greater experience for look gestures, lips shades, eye colors and other beauty products too. This reformes digital virtual face makeup with exact colour orientations. This is an important way to increment the sales and impress the clients in a fun way of new brand glance looks. The next company which offers this is Rolex which assists augmented reality to stimulate commercial needs and acquires purchase items. Here, the experience is determined based on the virtual phase of clients model and style.

(iv) AR in healthcare

Augmented reality is a powerful phase of healthcare and it mainly focuses on medical student technologies. This helps in training the environment of AR and enhances the telemedicine to transmit direct connection between professional and patients. At rigid conditions certain application of AR helps in motoring real-time data and assists in curing such as identifying and diagnosing human-body surgeries. The augmented reality in healthcare has introduced a technology device AccuVein which analyzes a patient's veins and guides with a lower escalation of 45% constantly. They have also enhanced a diagnostic tool on AR where the surgeons can determine the models and operation cuts in condition with tumor diseases. Further, research on AR Deloitte's cause lots of healthcare operations which increases the business model technically using the latest technology.

(v) AR for fun

Rather than the increase of the sales, most of the time augmented reality is used for fun purposes only. This assists in engaging the people with fun games such as teeth brushing, drawing and building blocks over AR technology. They have developed an app named DrawyBook which enhances all sketching skills. Proceeding from gaming to eastabilng AR browsers have lots of constructive ideas in deploying camera display. The augmented reality apps are introduced for growing up in an advanced industry pattern. In addition the ideology is to develop an Augmented reality touch for numerous lives.

## VII. CONCLUSION

The whole Scientists say that, once we are in the middle of the technology world, our brain will evolve to the next level, because the brain can't find the real or fake things happening there. Worst case is, if we jump over the building in AR we are still alive but in reality we died. That may happen in future if we handle spatial computers overly. Another term most of the classes and lectures will be in AR,

that is good news for all students and professors. Your home will be your classroom. Future is always unpredictable, that's why we are still seeking something important. Plus or minus in technologies but the world can't stop growing the technologies. The conclusion is that every technology has its beauty side and defect side but we must take all the importance into the count. Thus, spatial computing enables the technology of Augmented Reality (AR) in order to utilize and capture the information towards the physical and digital world.

#### *Future Works*

In general, AR doesn't include only in phones or tablets but it enhances various researches on touch lenses, wearable devices, voice control gadgets and many more. The latest intention of augmented reality is to determine appropriate and genuine involvement over phones or tablets but this can't be restored exactly. Even glasses might be replaced in the form of "smart glasses" but this mainly focuses on blind peoples. Similarly, all developing technology on AR has several diplomatic and social issues which instantly have higher privacy agitate over Google Glass too. Some peoples bothered more on recording chats and snapping images but this could be assessed based on face recognition application software. In production and sale augmented reality have greater demand over latest brand technologies like AR glasses, AR contacts, AR Glass-X and AR Google Lens. Finally, Augmented reality helps in enhancing the future over digitized real world. occupation generation.

### *References*

- [1] S. Balakrishnan, J. Janet, Aswin G. Artificial Intelligence and Robotics: A Research Overview, CSI Communications magazine, Volume No. 44, Issue No. 7, October 2020, pp. 24-26.
- [2] Balakrishnan. S, Artificial Intelligence in a Quantum World of Computation, CSI Communications magazine, Volume No. 44, Issue No. 3, June 2020, pp. 28-30.
- [3] Balakrishnan. S and K L Shunmuganathan. Article: A JADE Implementation of Integrated Agent System for E-Mail Coordination (IASEC). International Journal of Computer Applications 58(5): 5-9, November 2012.
- [4] S.Balakrishnan, "An Overview of Agent Based Intelligent Systems and Its Tools", CSI Communications magazine, Volume No. 42, Issue No. 10, January 2019, pp. 15-17.
- [5] S.Balakrishnan, "Agent based Bioinformatics Integration using JADE – Overview", CSI Communications magazine, Volume No. 43, Issue No. 9, December 2019, pp. 29-31.
- [6] Balakrishnan S and Steven Uaturomuinjo Tjiraso, "Integration of Agent Based Computing with Cloud Computing: Towards Cloud Intelligent Systems", International Research Publication House, Delhi. Engineering and Technology: Recent Innovations & Research, ISBN- 978-93-86138-06-4, pp. 1-17.
- [7] S. Balakrishnan, K.N. Sivabalan and J. Janet "MASFE - Multiagent System for Filtering E-Mails Using JADE", Advanced Engineering Research and Applications (AERA), Research India Publications, ISBN- 978-93-84443-42-9, pp. 148-167, 2017.
- [8] P.Arivazhagan, Balakrishnan. S and K L Shunmuganathan. "An Agent Based Centralized Router with Dynamic Connection Management Scheme Using JADE", International Journal of Applied Engineering Research, ISSN 0973-4562, Volume 11, Number 3 (2016) pp 2036-2041.
- [9] Manojkumar, D., P. G. Scholar, CV Arul Kumar, and Coimbatore Kinathukadavu. "Privacy Policy Multiparty Access Control On Content Sharing Sites.
- [10] Balakrishnan. S and K L Shunmuganathan, R. Sreenevasan, "Amelioration of Artificial Intelligence using Game Techniques for an Imperfect Information Board Game Geister" International Journal of Applied Engineering Research (IJAER). ISSN 0973-4562. Vol 9, Number 22 (2014) pp. 11849-11860.